

Circuit Simulation Lab:16

Design of Synchronous Circuits

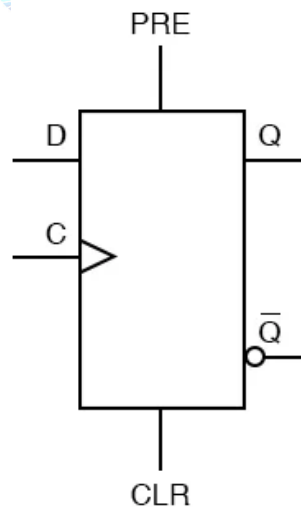
Introduction

The purpose of this experiment is to introduce the design of Synchronous sequential circuits, in this case Syn-D Flipflop. Synchronous inputs on a flip-flop have control over the outputs (Q and not-Q) only when the clock input has a change either rising edge or falling edge. These inputs are called the preset (PRE) and clear (CLR). The preset input drives the flip-flop to a set state while the clear input drives it to a reset state.

Software tools Requirement

Modelsim (Siemens)

Xilinx Vivado



Synchronous D Flip Flop

Describe the above circuits in Verilog HDL and capture the Waveforms

Behavior modeling